

Lab-8 (DMA)

1. Write a c-program to demonstrate DMA using.

i.malloc ii.calloc iii.realloc iv.free

Ans:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int i,*p,*q;
    p=(int *)malloc(4);
    q=(int *)calloc(5,4);

    printf("For malloc part:\n");
    printf("Enter amount:");
    scanf("%d",p);
    printf("The enter amount is:%d\n",*p);

    printf("\nFor calloc part:\n");
    printf("Enter element of array:");
    for(i=0; i<5; i++)
    {
        scanf("%d",q+i);
    }
    printf("The enter amount is:\n");
    for(i=0; i<5; i++)
    {
        printf("%d,",*(q+i));
    }
    printf("\n\nFor realloc part:\n");
    printf("The address before realloc:%d\n",p);
    p=(int *)realloc(p,100);
    printf("The address after realloc is:%d\n",p);

    free(p);
    free(q);
    return 0;
}
```

Output:

```
C:\Users\ayus\OneDrive\Desktop\New folder\C-program\lab8 DMA.exe
For malloc part:
Enter amount:23400012
The enter amount is:23400012

For calloc part:
Enter element of array:2 3 4 6 1
The enter amount is:
2,3,4,6,1,

For realloc part:
The address before realloc:12653600
The address after realloc is:12653664

-----
Process exited after 42.96 seconds with return value 0
Press any key to continue . . .
```

2. Write a c-program to find largest element using DMA.

Ans:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int i,max,*p;
    p=(int *)calloc(5,sizeof(int));
    printf("Enter an element for an array:\n");
    for(i=0; i<5; i++)
    {
        scanf("%d",p+i);
    }

    max=*p;
    for(i=0; i<5; i++)
    {
        if(max<*(p+i))
        {
            max=*(p+i);
        }
    }
    printf("The largest element is %d",max);
    return 0;
}
```

Output:

```
C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\lab8 DMA.exe
Enter an element for an array:
9
-1
4
5
6
The largest element is 9
-----
Process exited after 20.25 seconds with return value 0
Press any key to continue . . .
```